

HWK 2
ADVANCED MACHINE LEARNING
DATA 642

Objective: The homework objectives encompass topics in optimization, linear algebra, and sparse reconstruction algorithms. Firstly, students will employ Lagrange multipliers to derive closed-form solutions for under-determined and over-determined linear systems, showcasing their understanding of least squares optimization. Next, they will demonstrate the properties of the ℓ_2 norm, proving its validity as a true norm and establishing its convexity. Additionally, students will implement a Python script to generate sparse vectors and sensing matrices, applying LASSO regression for sparse reconstruction. Through repeated experiments, they will analyze the probability of correct reconstruction under different conditions, investigating the impact of matrix sparsity on reconstruction accuracy. This assignment integrates theoretical concepts with practical programming skills, emphasizing the application of mathematical principles to real-world problems in data analysis and optimization.

Exercise 1 (5 points)

(2.5 points) Show, using Lagrange multipliers, that the ℓ_2 minimizer

$$\begin{aligned} & \underset{\boldsymbol{\theta} \in \mathbb{R}^l}{\text{minimize}} && \|\boldsymbol{\theta}\|^2 \\ & \text{subject to} && \mathbf{x}_n^\top \boldsymbol{\theta} = y_n, \quad n = 1, \dots, N \end{aligned}$$

where $l > N$ (under-determined) and $\mathbf{X} \in \mathbb{R}^{N \times l}$ accepts the unique closed form solution

$$\hat{\boldsymbol{\theta}} = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top)^{-1} \mathbf{y}.$$

(2.5 points) Now, show that for the system $\mathbf{y} = \mathbf{X} \boldsymbol{\theta}$ with $\mathbf{X} \in \mathbb{R}^{N \times l}$ and $N > l$ (over-determined) the least squares solution is given by

$$\hat{\boldsymbol{\theta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}.$$

Exercise 2 (5points)

Prove that the ℓ_2 (Euclidean) norm is a true norm, i.e., it satisfies the four conditions that define a norm. Moreover, prove that any function that is a norm is also a convex function.

Exercise 3 (5 points)

Generate in Python a sparse vector $\boldsymbol{\theta} \in \mathbb{R}^l$, $l = 100$, with its first five components taking random values drawn from a normal distribution with mean zero and variance one and the rest being equal to zero. Build, also, a sensing matrix $\mathbf{X}$ with $N = 30$ rows having samples normally distributed, with mean zero and variance $1/N$, in order to get 30 observations based on the linear regression model $\mathbf{y} = \mathbf{X} \boldsymbol{\theta}$. Then perform the following tasks.

- (a) Use a LASSO implementation to reconstruct $\boldsymbol{\theta}$ from $\mathbf{y}$ and $\mathbf{X}$.

- (b) Repeat the experiment 100 times, with different realizations of $\mathbf{X}$, in order to compute the probability of correct reconstruction (assume the reconstruction is exact when $\|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\| < 10^{-8}$).
- (c) Repeat the same experiment (100 times) with matrices of the form

$$\mathbf{X}(i, j) = \begin{cases} +\sqrt{\frac{\sqrt{p}}{N}}, & \text{with probability } \frac{1}{2\sqrt{p}} \\ 0, & \text{with probability } 1 - \frac{1}{\sqrt{p}} \\ -\sqrt{\frac{\sqrt{p}}{N}}, & \text{with probability } \frac{1}{2\sqrt{p}} \end{cases}$$

for p equal to 1, 9, 25, 36, 64 (make sure that each row and each column of $\mathbf{X}$ has at least a nonzero component). Give an explanation why the probability of reconstruction falls as p increases (observe that both the sensing matrix and the unknown vector are sparse).